Determinants

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Introduction

Definition of a Determinant

The determinant of square matrix A is a scalar value (real number), denoted det(A), det A or $|A|$. It measures the amount by which a linear transformation changes the area of a figure.

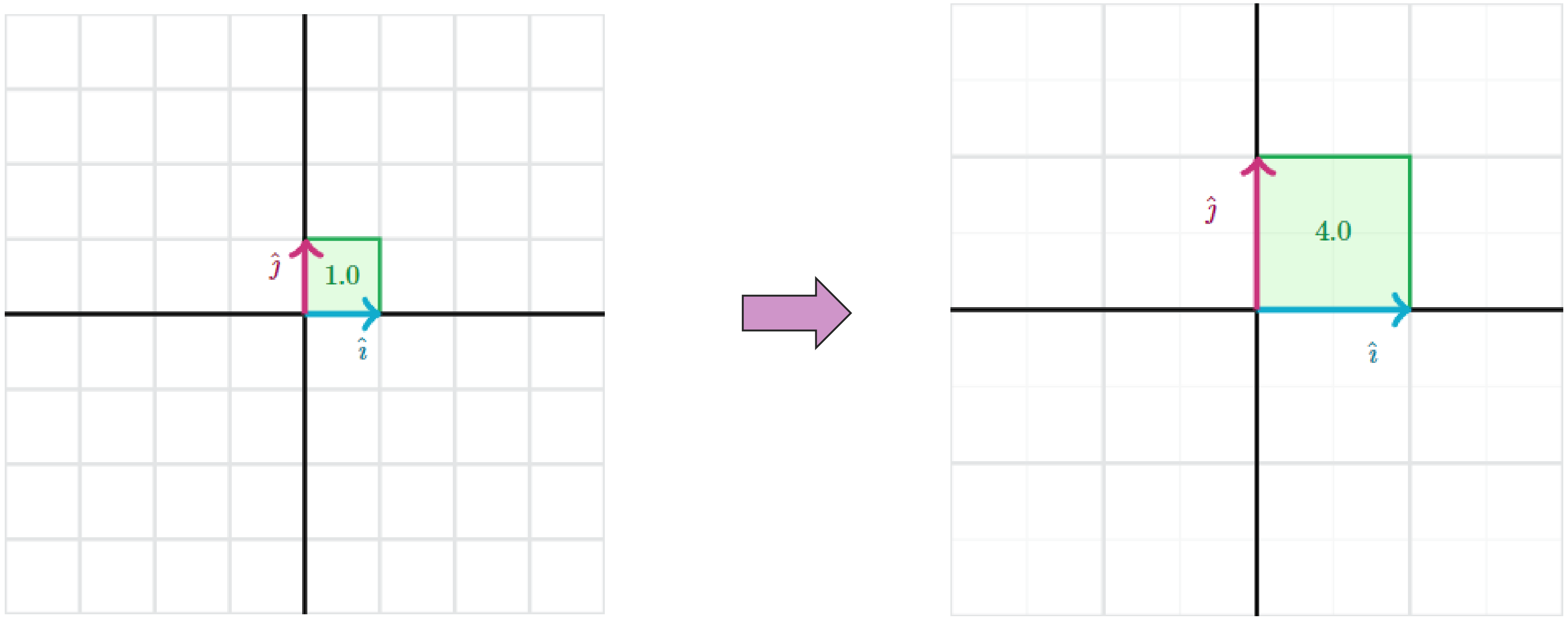
It can be considered as the scaling factor for the transformation of a matrix.

That is, when we use matrix transformations such as scaling, shearing, reflection, etc. the matrix stretches out the space or squeezes it in. The extent of this can be measured or represented as a determinant.

Introduction

Understanding Determinant

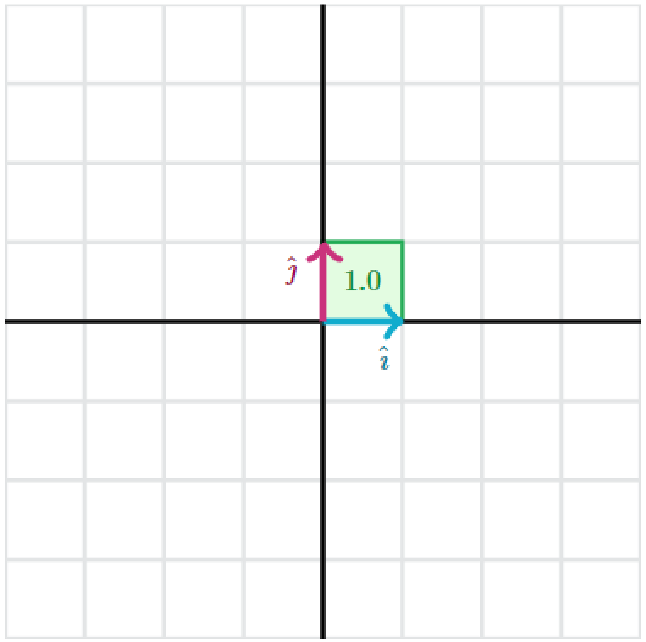
i. If a matrix stretches things out, then its determinant is greater than 1.

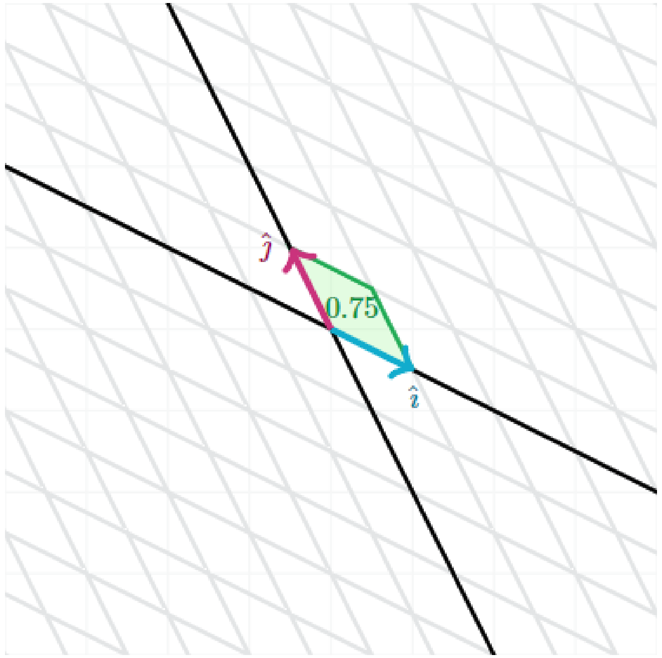


Introduction

Understanding Determinant

ii. If a matrix squeezes things in, then its determinant is less than 1.

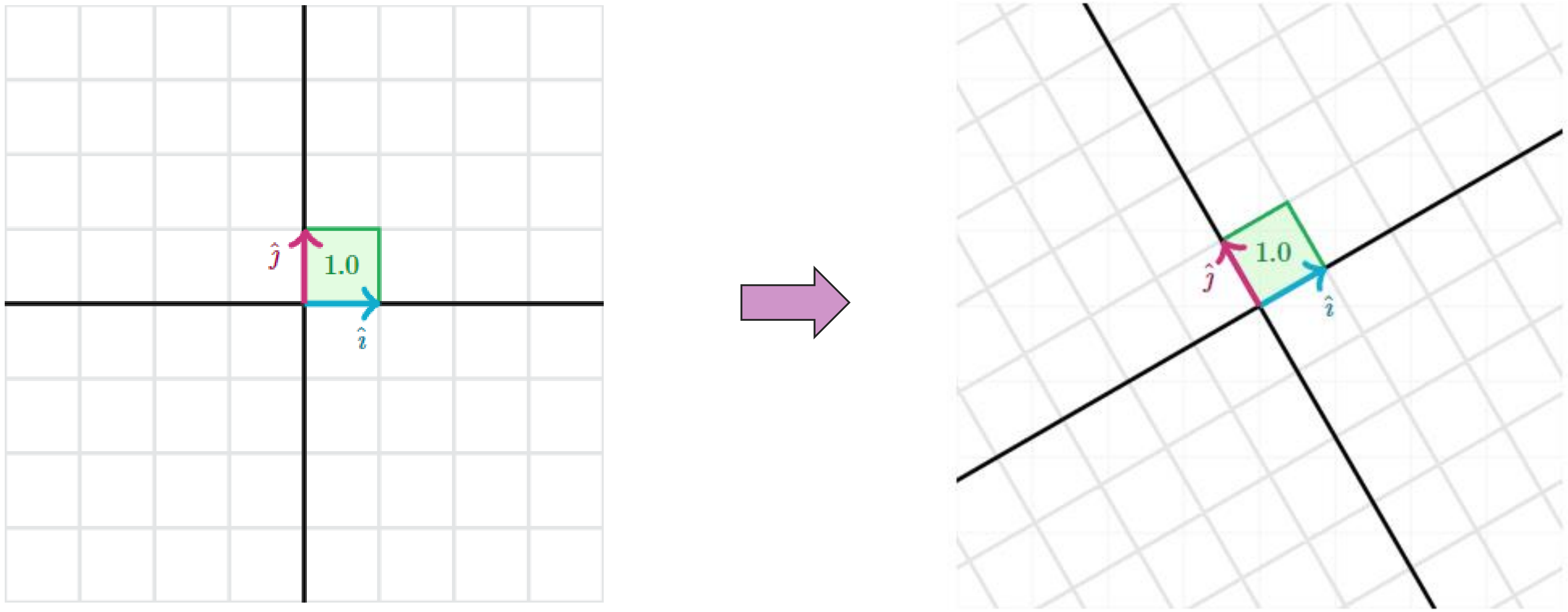




Introduction

Understanding Determinant

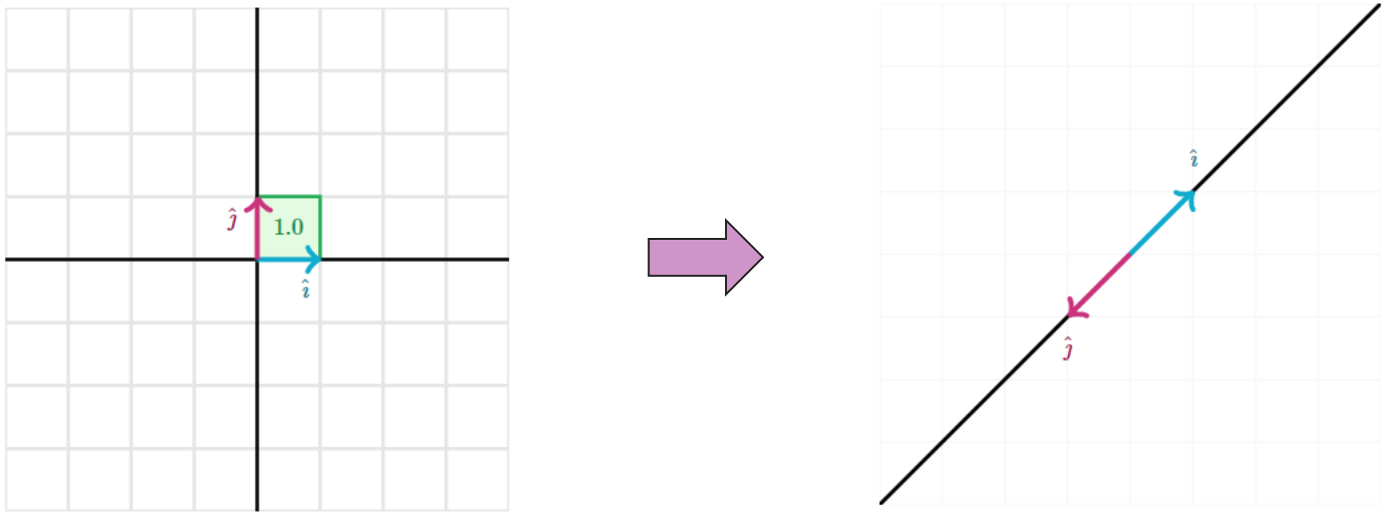
iii. If a matrix doesn't stretch things out or squeeze them in, then its determinant is exactly 1. An example of this is a rotation.



Introduction

Understanding Determinant

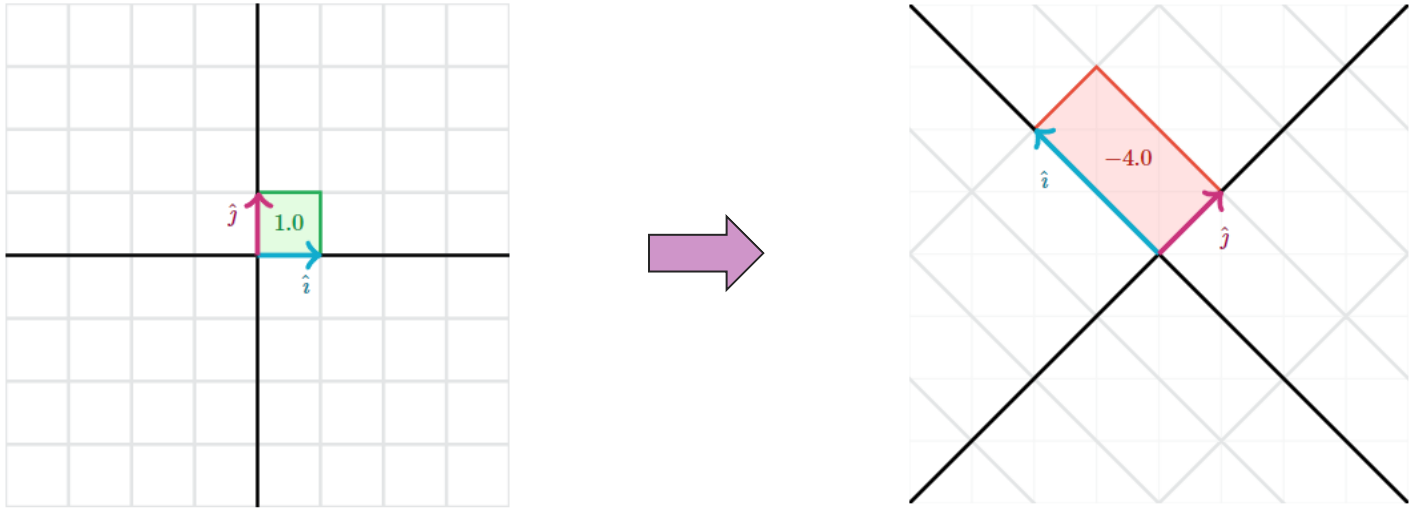
iv. Some matrices shrink space so much they actually flatten the entire grid on to a single line. This happens whenever a matrix maps the unit vectors $\hat{\imath}$ and $\hat{\jmath}$ to be multiples of each other, lying on the same line. These matrices have a determinant of 0.



Introduction

Understanding Determinant

v. Even though determinants represent scaling factors, they are not always positive numbers. The sign of the determinant has to do with the orientation of $\hat{\imath}$ and $\hat{\jmath}$. If a matrix flips the orientation, then its determinant is negative. Notice how $\hat{\imath}$ is on the left of $\hat{\jmath}$ in the image below, when normally it is on the right of $\hat{\jmath}$.



Introduction

Understanding Determinant

- The same idea of scaling area extends to 3D matrices as well. The only difference is that in 3D we say the matrix scales volume rather than area. The unit square also becomes the unit cube, whose sides are the unit vectors $\hat{\imath}$, $\hat{\jmath}$, and $\hat{k}$.
- One last important note is that the determinant only makes sense for square matrices. That's because square matrices move vectors from n-dimensional space to n-dimensional space, so we can talk about volume changing.

Determinants using Cofactor expansion

General Formula

The determinant of a 2×2 matrix is

$$\begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad - bc,$$

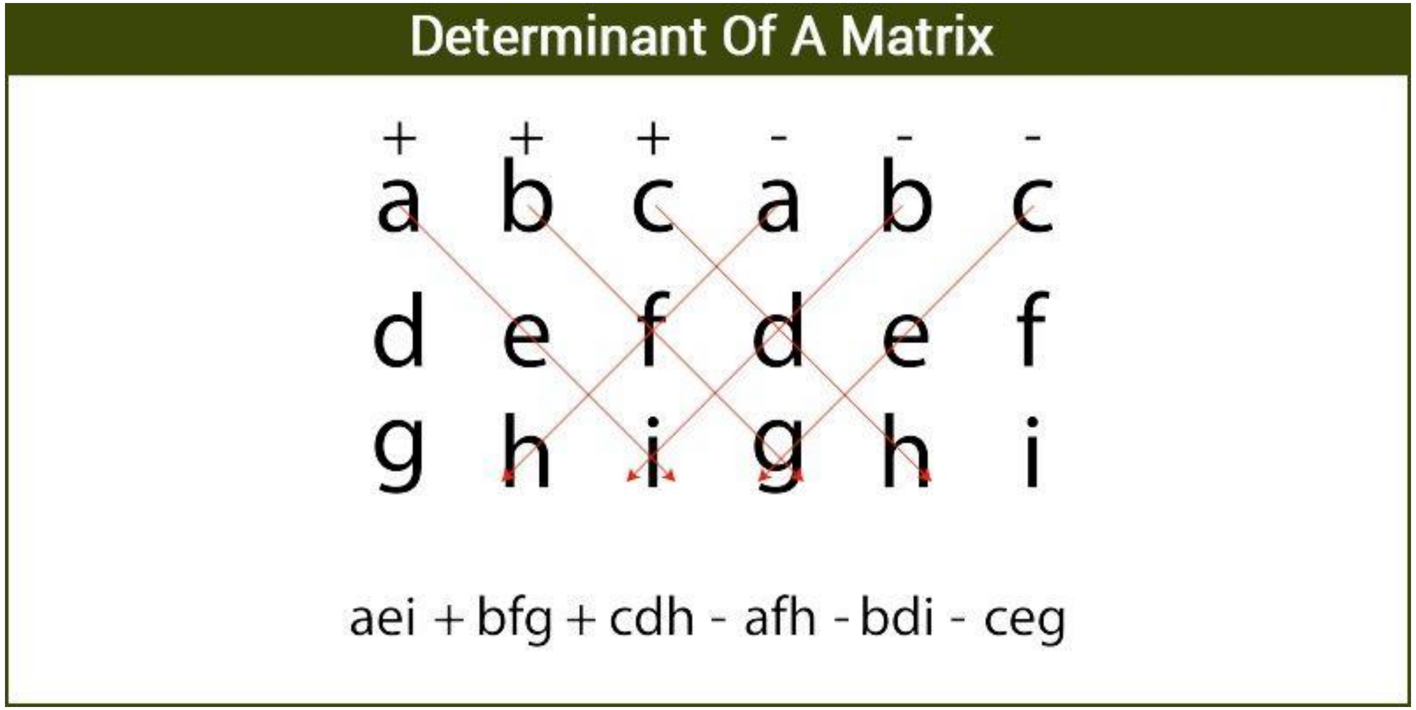
and the determinant of a 3×3 matrix is

$$\begin{vmatrix} a & b & c \\ d & e & f \\ g & h & i \end{vmatrix} = aei + bfg + cdh - ceg - bdi - afh.$$

Determinants using Cofactor expansion

General Formula

- Easier way to remember the determinant of a 3 x 3 matrix.



Determinants using Cofactor expansion

For any square matrix A, let A_{ij} denote the submatrix formed by deleting the ith row and jth column of A. For instance, if

$$A = \begin{bmatrix} 1 & -2 & 5 & 0 \\ 2 & 0 & 4 & -1 \\ 3 & 1 & 0 & 7 \\ 0 & 4 & -2 & 0 \end{bmatrix}$$

then A_{32} is obtained by crossing out row 3 and column 2,

$$\begin{bmatrix} 1 & -2 & 5 & 0 \\ 2 & 0 & 4 & -1 \\ 3 & 1 & 0 & 7 \\ 0 & 4 & -2 & 0 \end{bmatrix} \Longrightarrow \quad A_{32} = \begin{bmatrix} 1 & 5 & 0 \\ 2 & 4 & -1 \\ 0 & -2 & 0 \end{bmatrix}$$

Determinants using Cofactor expansion

Minor and Cofactor

Definition. Let A be an $n \times n$ matrix.

1. The (i, j) ***minor,*** denoted A_{ij}, is the $(n-1) \times (n-1)$ matrix obtained from A by deleting the ith row and the jth column.
2. The (i, j) ***cofactor*** C_{ij} is defined in terms of the minor by

$$C_{ij} = (-1)^{i+j} \det(A_{ij}).$$

Determinants using Cofactor expansion

EXAMPLE

For

$$A = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{pmatrix},$$

compute A_{23} and C_{23}.

Solution

$$A_{23} = \begin{pmatrix} 1 & 2 & \cancel{3} \\ \cancel{4} & \cancel{5} & \cancel{6} \\ 7 & 8 & \cancel{9} \end{pmatrix} = \begin{pmatrix} 1 & 2 \\ 7 & 8 \end{pmatrix} \qquad C_{23} = (-1)^{2+3} \det \begin{pmatrix} 1 & 2 \\ 7 & 8 \end{pmatrix} = (-1)(-6) = 6$$

Determinants using Cofactor expansion

We can now give a *recursive* definition of a determinant. When $n = 3$, $\det A$ is defined using determinants of the 2×2 submatrices A_{1j}, as in (3) above. When $n = 4$, $\det A$ uses determinants of the 3×3 submatrices A_{1j}. In general, an $n \times n$ determinant is defined by determinants of $(n-1) \times (n-1)$ submatrices.

For $n \geq 2$, the **determinant** of an $n \times n$ matrix $A = [a_{ij}]$ is the sum of n terms of the form $\pm a_{1j} \det A_{1j}$, with plus and minus signs alternating, where the entries $a_{11}, a_{12}, \dots, a_{1n}$ are from the first row of A. In symbols,

$$\begin{aligned} \det A &= a_{11} \det A_{11} - a_{12} \det A_{12} + \cdots + (-1)^{1+n} a_{1n} \det A_{1n} \\ &= \sum_{j=1}^{n} (-1)^{1+j} a_{1j} \det A_{1j} \end{aligned}$$

Determinants using Cofactor expansion

EXAMPLE Compute the determinant of $A = \begin{bmatrix} 1 & 5 & 0 \\ 2 & 4 & -1 \\ 0 & -2 & 0 \end{bmatrix}$

SOLUTION Compute $\det A = a_{11} \det A_{11} - a_{12} \det A_{12} + a_{13} \det A_{13}$:

$$\det A = 1 \cdot \det \begin{bmatrix} 4 & -1 \\ -2 & 0 \end{bmatrix} - 5 \cdot \det \begin{bmatrix} 2 & -1 \\ 0 & 0 \end{bmatrix} + 0 \cdot \det \begin{bmatrix} 2 & 4 \\ 0 & -2 \end{bmatrix}$$
$$= 1(0-2) - 5(0-0) + 0(-4-0) = -2$$

Note: Another common notation for the determinant of a matrix uses a pair of vertical lines in place of brackets. Thus the calculation in Example 1 can be written as

$$\det A = 1 \begin{vmatrix} 4 & -1 \\ -2 & 0 \end{vmatrix} - 5 \begin{vmatrix} 2 & -1 \\ 0 & 0 \end{vmatrix} + 0 \begin{vmatrix} 2 & 4 \\ 0 & -2 \end{vmatrix} = \cdots = -2$$

Determinants using Cofactor expansion

EXAMPLE Use a cofactor expansion across the third row to compute det A, where

$$A = \begin{bmatrix} 1 & 5 & 0 \\ 2 & 4 & -1 \\ 0 & -2 & 0 \end{bmatrix}$$

SOLUTION Compute

$$\begin{aligned} \det A &= a_{31}C_{31} + a_{32}C_{32} + a_{33}C_{33} \\ &= (-1)^{3+1}a_{31} \det A_{31} + (-1)^{3+2}a_{32} \det A_{32} + (-1)^{3+3}a_{33} \det A_{33} \\ &= 0\begin{vmatrix} 5 & 0 \\ 4 & -1 \end{vmatrix} - (-2)\begin{vmatrix} 1 & 0 \\ 2 & -1 \end{vmatrix} + 0\begin{vmatrix} 1 & 5 \\ 2 & 4 \end{vmatrix} \\ &= 0 + 2(-1) + 0 = -2 \end{aligned}$$

Determinants using Cofactor expansion

EXAMPLE Compute det A, where

$$A = \begin{bmatrix} 3 & -7 & 8 & 9 & -6 \\ 0 & 2 & -5 & 7 & 3 \\ 0 & 0 & 1 & 5 & 0 \\ 0 & 0 & 2 & 4 & -1 \\ 0 & 0 & 0 & -2 & 0 \end{bmatrix}$$

SOLUTION The cofactor expansion down the first column of A has all terms equal to zero except the first. Thus

$$\det A = 3 \cdot \begin{vmatrix} 2 & -5 & 7 & 3 \\ 0 & 1 & 5 & 0 \\ 0 & 2 & 4 & -1 \\ 0 & 0 & -2 & 0 \end{vmatrix} - 0 \cdot C_{21} + 0 \cdot C_{31} - 0 \cdot C_{41} + 0 \cdot C_{51}$$

Determinants using Cofactor expansion

Theorem (Cofactor expansion). *Let A be an $n \times n$ matrix with entries a_{ij}.*

1. *For any $i = 1, 2, \ldots, n$, we have*

$$\boxed{\det(A) = \sum_{j=1}^{n} a_{ij} C_{ij} = a_{i1} C_{i1} + a_{i2} C_{i2} + \cdots + a_{in} C_{in}.}$$

*This is called **cofactor expansion along the ith row.***

2. *For any $j = 1, 2, \ldots, n$, we have*

$$\boxed{\det(A) = \sum_{i=1}^{n} a_{ij} C_{ij} = a_{1j} C_{1j} + a_{2j} C_{2j} + \cdots + a_{nj} C_{nj}.}$$

*This is called **cofactor expansion along the jth column.***

Determinants using Cofactor expansion

Recipe: Computing the Determinant of a 3×3 Matrix.

To compute the determinant of a 3×3 matrix, first draw a larger matrix with the first two columns repeated on the right. Then add the products of the downward diagonals together, and subtract the products of the upward diagonals:

$$\det\begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix} = \begin{matrix} a_{11}a_{22}a_{33} + a_{12}a_{23}a_{31} + a_{13}a_{21}a_{32} \\ -a_{13}a_{22}a_{31} - a_{11}a_{23}a_{32} - a_{12}a_{21}a_{33} \end{matrix}$$

$$\begin{matrix} a_{11} & a_{12} & a_{13} & a_{11} & a_{12} \\ a_{21} & a_{22} & a_{23} & a_{21} & a_{22} \\ a_{31} & a_{32} & a_{33} & a_{31} & a_{32} \end{matrix} \quad - \quad \begin{matrix} a_{11} & a_{12} & a_{13} & a_{11} & a_{12} \\ a_{21} & a_{22} & a_{23} & a_{21} & a_{22} \\ a_{31} & a_{32} & a_{33} & a_{31} & a_{32} \end{matrix}$$

Determinants using Cofactor expansion

Choosing favorable rows or columns to calculate Determinant

Cofactor expansions are most useful when computing the determinant of a matrix that has a row or column with several zero entries. Indeed, if the (i, j) entry of A is zero, then there is no reason to compute the (i, j) cofactor. In the following example we compute the determinant of a matrix with two zeros in the fourth column by expanding cofactors along the fourth column.

Determinants using Cofactor expansion

EXAMPLE Find the determinant of

$$A = \begin{pmatrix} 2 & 5 & -3 & -2 \\ -2 & -3 & 2 & -5 \\ 1 & 3 & -2 & 0 \\ -1 & 6 & 4 & 0 \end{pmatrix}.$$

Solution

The fourth column has two zero entries. We expand along the fourth column to find

$$\det(A) = 2\det\begin{pmatrix} -2 & -3 & 2 \\ 1 & 3 & -2 \\ -1 & 6 & 4 \end{pmatrix} - 5\det\begin{pmatrix} 2 & 5 & -3 \\ 1 & 3 & -2 \\ -1 & 6 & 4 \end{pmatrix}$$
$$- 0\det(\text{don't care}) + 0\det(\text{don't care}).$$

Determinants using Cofactor expansion

SOLUTION (Continued...)

We only have to compute two cofactors. We can find these determinants using any method we wish; for the sake of illustration, we will expand cofactors on one and use the formula for the 3×3 determinant on the other.

Expanding along the first column, we compute

$$
\begin{aligned}
&\det\begin{pmatrix} -2 & -3 & 2 \\ 1 & 3 & -2 \\ -1 & 6 & 4 \end{pmatrix} \\
&= -2\det\begin{pmatrix} 3 & -2 \\ 6 & 4 \end{pmatrix} - \det\begin{pmatrix} -3 & 2 \\ 6 & 4 \end{pmatrix} - \det\begin{pmatrix} -3 & 2 \\ 3 & -2 \end{pmatrix} \\
&= -2(24) - (-24) - 0 = -48 + 24 + 0 = -24.
\end{aligned}
$$

Determinants using Row Reduction

- A matrix can always be transformed into row echelon form by a series of row operations, and a matrix in row echelon form is upper-triangular.

No matter which row operations you do, you will always compute the same value for the determinant.

Determinants using Row Reduction

Properties of Determinants

Determinants can also be defined by some of their properties. Namely, the determinant is the unique function defined on the $n \times n$ matrices that has the four following properties:

1. Doing a row replacement on A does not change $\det(A)$.
2. Scaling a row of A by a scalar c multiplies the determinant by c.
3. Swapping two rows of a matrix multiplies the determinant by -1.
4. The determinant of the identity matrix I_n is equal to 1.

Determinants using Row Reduction

EXAMPLE Compute $\det\begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 5 & 1 & 0 \end{pmatrix}$.

Solution

First we row reduce, then we compute the determinant in the opposite order:

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 5 & 1 & 0 \end{pmatrix} \qquad \det = -1$$

$$\xrightarrow{R_2 \longleftrightarrow R_3} \begin{pmatrix} 1 & 0 & 0 \\ 5 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \qquad \det = 1$$

$$\xrightarrow{R_2 = R_2 - 5R_1} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \qquad \det = 1$$

Determinants using Row Reduction

SOLUTION (Continued...)

The reduced row echelon form is I_3, which has determinant 1. Working backwards from I_3 and using the four defining properties, we see that the second matrix also has determinant 1 (it differs from I_3 by a row replacement), and the first matrix has determinant -1 (it differs from the second by a row swap).

Determinants using Row Reduction

Here is the general method for computing determinants using row reduction.

Recipe: Computing determinants by row reducing.

Let A be a square matrix. Suppose that you do some number of row operations on A to obtain a matrix B in row echelon form. Then

$$\det(A) = (-1)^r \cdot \frac{(\text{product of the diagonal entries of } B)}{(\text{product of scaling factors used})},$$

where r is the number of row swaps performed.

Determinants using Row Reduction

Example (The determinant of a 2×2 matrix). Let us use the recipe to compute the determinant of a general 2×2 matrix $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$.

$$\boxed{\det\begin{pmatrix} a & b \\ c & d \end{pmatrix} = ad - bc.}$$

- If $a = 0$, then

$$\det\begin{pmatrix} a & b \\ c & d \end{pmatrix} = \det\begin{pmatrix} 0 & b \\ c & d \end{pmatrix} = -\det\begin{pmatrix} c & d \\ 0 & b \end{pmatrix} = -bc.$$

- If $a \neq 0$, then

$$\begin{aligned}\det\begin{pmatrix} a & b \\ c & d \end{pmatrix} &= a \cdot \det\begin{pmatrix} 1 & b/a \\ c & d \end{pmatrix} = a \cdot \det\begin{pmatrix} 1 & b/a \\ 0 & d - c \cdot b/a \end{pmatrix} \\ &= a \cdot 1 \cdot (d - bc/a) = ad - bc.\end{aligned}$$

Determinants using Row Reduction

EXAMPLE Compute $\det \begin{pmatrix} 0 & -7 & -4 \\ 2 & 4 & 6 \\ 3 & 7 & -1 \end{pmatrix}$.

Solution

We row reduce the matrix, keeping track of the number of row swaps and of the scaling factors used.

$$\begin{pmatrix} 0 & -7 & -4 \\ 2 & 4 & 6 \\ 3 & 7 & -1 \end{pmatrix} \xrightarrow{R_1 \longleftrightarrow R_2} \begin{pmatrix} 2 & 4 & 6 \\ 0 & -7 & -4 \\ 3 & 7 & -1 \end{pmatrix} \qquad r = 1$$

$$\xrightarrow{R_1 = R_1 \div 2} \begin{pmatrix} 1 & 2 & 3 \\ 0 & -7 & -4 \\ 3 & 7 & -1 \end{pmatrix} \qquad \text{scaling factors} = \frac{1}{2}$$

$$\xrightarrow{R_3 = R_3 - 3R_1} \begin{pmatrix} 1 & 2 & 3 \\ 0 & -7 & -4 \\ 0 & 1 & -10 \end{pmatrix}$$

Determinants using Row Reduction

SOLUTION (Continued...)

$$\xrightarrow{R_2 \longleftrightarrow R_3} \begin{pmatrix} 1 & 2 & 3 \\ 0 & 1 & -10 \\ 0 & -7 & -4 \end{pmatrix} \quad r = 2$$

$$\xrightarrow{R_3 = R_3 + 7R_2} \begin{pmatrix} 1 & 2 & 3 \\ 0 & 1 & -10 \\ 0 & 0 & -74 \end{pmatrix}$$

We made two row swaps and scaled once by a factor of $1/2$, so the recipe says that

$$\det\begin{pmatrix} 0 & -7 & -4 \\ 2 & 4 & 6 \\ 3 & 7 & -1 \end{pmatrix} = (-1)^2 \cdot \frac{1 \cdot 1 \cdot (-74)}{1/2} = -148.$$

Determinants using Row Reduction

EXAMPLE Compute $\det\begin{pmatrix}1 & 2 & 3\\ 2 & -1 & 1\\ 3 & 0 & 1\end{pmatrix}$.

Solution

We row reduce the matrix, keeping track of the number of row swaps and of the scaling factors used.

$$\begin{pmatrix}1 & 2 & 3\\ 2 & -1 & 1\\ 3 & 0 & 1\end{pmatrix} \xrightarrow[R_3=R_3-3R_1]{R_2=R_2-2R_1} \begin{pmatrix}1 & 2 & 3\\ 0 & -5 & -5\\ 0 & -6 & -8\end{pmatrix}$$

$$\xrightarrow{R_2=R_2\div -5} \begin{pmatrix}1 & 2 & 3\\ 0 & 1 & 1\\ 0 & -6 & -8\end{pmatrix} \quad \text{scaling factors} = -\tfrac{1}{5}$$

$$\xrightarrow{R_3=R_3+6R_2} \begin{pmatrix}1 & 2 & 3\\ 0 & 1 & 1\\ 0 & 0 & -2\end{pmatrix}$$

Determinants using Row Reduction

SOLUTION (Continued...)

We did not make any row swaps, and we scaled once by a factor of $-1/5$, so the recipe says that

$$\det\begin{pmatrix} 1 & 2 & 3 \\ 2 & -1 & 1 \\ 3 & 0 & 1 \end{pmatrix} = \frac{1 \cdot 1 \cdot (-2)}{-1/5} = 10.$$